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ENVIRONMENTAL PROGRESS & SUSTAINABLE ENERGY

ORIGINAL RESEARCH

Sustainable Energy

Experimental analysis of earth air heat exchanger for space cooling: A clean energy approach

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Abstract

In response to the rising energy demands and increasing reliance on air conditioning due to extreme heat in India, this study attempts to investigate the thermal performance and sustainability of a horizontally configured Earth Air Heat Exchanger (EAHE) system. Thermal modeling was developed using MATLAB R2024b and has been experimentally validated for a 4 ft. × 4 ft. × 4 ft. room. This setup was installed in Dadri, India and experimentally tested for 3 days under varying air flow velocities. The system demonstrated significant cooling potential, with temperature drop of up to 7.3°C between the inlet and outlet air of EAHE. The correlation coefficient ($r = 0.84$ to 0.89) and root mean square percent deviation (<4%) between the theoretical and experimental values confirms the reliability of the model. The validated theoretical model was then applied to a realistic sized room of 12 ft × 12 ft × 12 ft, showing an average reduction of 480 W in cooling load of room which corresponded to a seasonal energy saving of 468 kWh. The Energy Payback Time (EPBT) of EAHE was calculated to be 4.42 years and a potential carbon mitigation of 43.5 ton over 50 years and a carbon credit value of $435. This study highlights the potential of EAHE as a sustainable, low energy solution for passive cooling in India, resulting in reduced electricity consumption and carbon emissions which is aligned with UN goals namely, SDG 7, SDG 11, SDG12, and SDG 13.

KEYWORDS
carbon credit, correlation coefficient, earth air heat exchanger, energy payback time, thermal modeling

1 | INTRODUCTION

As the energy demand of the world grew, more non-renewable sources of energy were used without considering their impact on the environment. Consequently, global warming increased. Hot days turned hotter. Like a domino effect, energy demand increased exponentially. So is the case in India. According to a report by the Ministry of Statistics and programme implementation (year), the annual electricity consumption per capita in India stands at 1015 kWh, which corresponds to approximately two tons of carbon emissions per capita if the electricity were produced

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from coal. Moreover, being a developing nation, it heavily relies on coal for power generation. Rising temperatures and frequent heat waves in India have intensified electricity demands, driven primarily by air conditioners. Space cooling accounts for nearly 10% of a country's electricity demand, growing by 21% from 2019 to 2022. The International Energy Agency (IEA)[1] estimates that by 2050, space cooling will account for 45% of the country's electricity demand. Hence, there is a need for innovative solutions that integrate renewable sources of energy such as solar energy, wind energy, and geothermal energy within the HVAC systems to reduce their load requirements and therefore, reduce carbon emissions.

Bisoniya[2] proposed that the earth–air heat exchanger is a promising technique which can effectively be used to preheat the air in winter and vice versa in summer. He found that a longer pipe of smaller diameter buried at a greater depth and having lower air flow velocity results in an increase in performance of the Earth-to-Air Heat Exchanger (EAHE) system. Ahmed et al.[3] did the parametric study on thermal performance of horizontal earth pipe cooling system in summer for the hot humid subtropical climate at Rockhampton, Australia with the help of a thermal model developed in simulation program, FLUENT 15.0. Results showed a significant effect for air velocity, pipe length, and pipe diameter on the earth pipe cooling performance, where the pipe length dominated the other parameters.

Mohammed et al.[4] presented a MATLAB/Simulink model in order to predict the distribution of soil temperatures and design EAHEs efficiently in arid locations. The study proved that the installation of the EAHEs in arid locations like Kufa, Iraq, is highly efficient, achieving a cooling potential of 1626 W in August and 129.8 W in March. Efanden et al.[5] used a control volume method which is implemented by dividing the earth air heat exchanger duct into identical microvolumes and then establishing an energy balance for each microvolume. This method was used to optimize the total length of the duct of a ground air heat exchanger installed in an equatorial zone which was found to be 40 m. El-Said et al.[6] established numerical modeling to predict the effects of different working parameters on the aerothermal performance of EAHE in summer conditions. It found that the performance of EAHE enhances by increasing the buried pipe length and vice versa with the Reynolds number increase.

Xiao et al.[7] investigated the influence of different types of pipes (aluminum, stainless steel, polyvinyl chloride, corrugated, and perforated corrugated pipes) on the performance of EAHE. Corrugated pipe had the best overall performance from which it can be concluded that pipe with rough surfaces was the most suitable pipe type for EAHE.

Molina-Rodea et al.[8] evaluated a vertical heat exchanger connected in series and found that a maximum COP of 12.8 and a maximum effectiveness of 88.4% are achieved, which are sufficient for cooling purposes in areas with space restrictions. Taşdelen et al.[9] applied computational fluid dynamics (CFD) in ANSYS FLUENT 19.2 software to analyze the cooling performance of 3D-modeled different types (4 horizontal and 4 vertical) EAHEs with equal pipe lengths and found that the vertical EAHE produced more thermal energy than the horizontal EAHE. Sofyan et al.[10] used ANSYS Fluent to compare the thermal performance of series and parallel EAHE for cooling and found that the parallel EAHE gives a higher temperature drop and is 15% more effective than the series EAHE by using the effectiveness-NTU relationship. Qi et al.[11] introduces an innovative design for EAHE utilizing a brachistochrone pipe configuration. Compared to the ordinary EAHE, this configuration increases the heat transfer rate by 5.33%.

Sghiouri et al.[12] developed a framework in which the optimization focuses on minimizing the life cycle cost (LCC) and maximizing the cooling potential across three distinct Moroccan climates. Yue et al.[13] developed a multi-objective framework considering heat exchange efficiency, average thermal capacity, coefficient of performance, and payback time as objective functions. One of the optimal solutions based on the entropy weight method obtained 75.4% heat exchange efficiency, 727.4 W average thermal capacity, a coefficient of performance of 20.4, and a 5.28-year payback time. Yue et al.[14] deployed machine learning and developed a machine learning-based multi-performance prediction model. Out of the models made, the optimized-extreme gradient boosting model performed the best, with a determination coefficient of 0.98, root mean squared error of 22.98, and mean absolute error of 8.88.

According to Bady et al.,[15] the continuous operation of the EAHE for a long time leads to ground saturation with heat, which means that the ground cannot absorb more heat from the EAHE inlet air. Therefore, Bady et al.[15] investigates the performance improvement of EAHE by applying a control strategy to the EAHE system according to two different modes where mode 1 is applied when the temperature difference between the ambient air and soil exceeds 1.5°C and mode 2 is applied when it does not satisfy mode 1 and the ambient air is fed directly to the room. Jiang et al.[16] highlights the significance of considering the soil freezing effect in cold regions for EAHE design. Otherwise, the calculated soil temperature and EAHE outlet air temperature will be too low, resulting in significant deviations from the actual situation. Qi et al.[17] proposed a more durable EAHE with a return air system to alleviate thermal saturation of soil.

Patin et al.[18] conduct an economic viability analysis for EAHE focusing on a typical single-family residence in Canada and found that the optimal EAHE configuration reduces the building's heating/cooling load by 701 kWh per year. Moreover, they determined that only 60% of the heat exchanged with the soil effectively contributes to load reduction.

Dhaidan et al.[19] investigated the integration of EAHE in the solar greenhouse system to provide the necessary heating and cooling required to maintain suitable conditions for vegetation. Jilani et al.[20] investigated a greenhouse integrated with thin-film photovoltaics and EAHE and found that the greenhouse can save 41,587 kWh/year after integrating the EAHE. Also, the air temperature in the greenhouse room rises by 5°C in the winter and falls by 17°C in the summer due to the integration of EAHE. Ren et al.[21] explored phase change material—filled earth-air heat exchanger (PAHE) using the finite difference method, with a computational program implemented on MATLAB. They achieved a maximum temperature drop of 5.26°C during the nine-day investigation period. Shahsavar et al.[22] investigated the integration of EAHE with a photovoltaic system equipped with a phase changing material. They found that the optimal system can fulfill around 133 MWh of heating demand for the building during cold months, address 27 MWh of cooling needs during hot months, and generate 9 MWh electrical power throughout the entire year.

The literature reviews illustrate that EAHEs are an effective and energy-efficient solution for heating and cooling across diverse climatic conditions. The performance of EAHEs is strongly affected by design and operating parameters such as pipe length, pipe diameter, pipe material, burial depth, and airflow velocity. Longer pipes, smaller diameters, larger burial depths, and lower airflow velocities generally enhance the thermal performance of EAHE. Length of the pipe is found to be the dominating parameter among various parameters. Simulations and CFD results confirm that vertical and parallel EAHE configurations can perform better as compared with conventional horizontal and series configurations. Corrugated pipes improve heat transfer, and innovative designs such as brachistochrone pipes and phase-change-material-assisted EAHEs further increase temperature drop and thermal capacity. Climate-specific studies highlight high cooling potential in dry and humid climates, while cold-region applications require consideration of soil freezing effects and thermal saturation. To address long-term ground saturation, control strategies and return-air systems have been proposed, significantly improving system sustainability. Recent research has expanded toward optimization, life-cycle cost analysis, and multi-objective frameworks incorporating performance, economic viability, and payback time. Moreover, machine learning models—particularly gradient boosting techniques—have shown excellent accuracy in predicting EAHE performance. Integration of EAHE with greenhouses, photovoltaic systems, and solar applications demonstrates substantial energy savings and improved indoor thermal conditions, confirming EAHEs as a versatile and promising technology for sustainable building energy systems.

In this paper, the EAHE is designed to utilize the Earth's stable underground temperature to preheat or precool the air which can reduce the load of heaters during winter and air conditioners during summers. The earth temperature beyond a depth of 1 meter is usually insensitive to the diurnal cycle of air temperature and solar radiation.[23] Different theoretical methods investigated using various software and validated to optimize the EAHE system so that maximum amount of heat transfer obtain. A horizontal pipe configured EAHE is investigated experimentally for 3 days with varying air flow velocity and further validated theoretically. This study uses LMTD method which is implemented on MATLAB R2024b to predict the exit air temperature and evaluate the thermal energy. Moreover, analysis of the EAHE on 4 ft $\times$ 4 ft $\times$ 4 ft room is done experimentally and further validated theoretically. Again, the thermal modeling of the Room + EAHE system is implemented on MATLAB R2024b to predict the temperature of the room. Moreover, the total embodied energy for the system, energy payback time, CO_2 mitigation and carbon credit earned are evaluated.

2 | EXPERIMENTAL SETUP

2.1 | Materials and methods

In this study, a horizontal configured EAHE made of PVC material has been used to precool a cubical room made of concrete 4 ft $\times$ 4 ft $\times$ 4 ft walls with each vertical walls facing north, east, west and south directions as shown in Figure 1. The ground floor of the room has been considered adiabatic and hence, no heat transfer occurs through the floor. The PVC pipe has been buried at a depth of 6 feet having pipe length 15 feet and pipe diameter of 10 inches.

The outside ambient air is blown into the heat exchanger through a 550 W plastic blower with air velocity of 0.5, 1, and 1.2 m/s. This velocity is varied by changing the nozzle of the blower, that is, a diverging nozzle to decrease the velocity or a converging nozzle to increase the velocity. To measure the temperatures of air at inlet and outlet of the heat exchanger as well as to measure the wall temperatures of the room, K type thermocouples are used. These thermocouples are connected to a data logger to save the values of the temperature every 5 min interval. A mercury thermometer was used to measure the ambient temperature and room temperature every hour. A digital anemometer is used to measure the velocity of air at the exit of the heat exchanger. The setup at Shiv Nadar University, near New Delhi, India (28°36′N 77°12′E). The reading of the azimuth angle of the sun and the solar radiation incident on the room (New Delhi) were taken from IMD (Indian Meteorological Department) Pune. The experiment was done on March 22, April 3 and April 4 (2015) with air velocity of 0.5, 1, and 1.2 m/s, respectively.

3 | METHODOLOGY

The methodology adopted for the study is shown in the figure listed below. The data of solar radiation and azimuth angle for New Delhi is taken from IMD Pune. All other data used as input to theoretical modeling in MATLAB is derived from the experimental setup. Subsequently, the theoretical model is validated by experimental findings by calculating the correlation coefficient and the root mean square deviation.

1. Thermal Modeling of the EAHE is developed in MATLAB 2024b.
2. Using the ambient inlet air temperature measurements from thermocouple, outlet air temperature and rate of heat transfer are obtained from the MATLAB model.
3. The value of exit temperature obtained from the MATLAB modeling is validated with the experimental findings by calculating the correlation coefficient and root mean square percent deviation.
4. The data of solar radiation and the azimuth angle is taken from IMD Pune and is converted to get the solar radiation incident on 90° vertical walls in north, south, east and west directions and horizontal flat roof and is shown in Table 1
5. The energy balance equation for the room and EAHE is modeled in MATLAB 2024b and MATLAB Simulink, which gives the temperature of the room.
6. The value of room temperature obtained from the MATLAB modeling is validated with the experimental findings by calculating the correlation coefficient and root mean square percent deviation.
7. The reduction in cooling load for an air conditioner is calculated for a realistic size of room 12 ft $\times$ 12 ft $\times$ 12 ft (8.72 m $\times$ 8.72 m $\times$ 8.72 m) once the theoretical model is validated.



FIGURE 1 Photographs of the constructed EAHE system. (a) The complete EAHE system. (b) The PVC pipe buried in the ground. (c) Data Logger. (d) Plastic blower of 550 W.

TABLE 1 Solar Radiation incident on various walls of the room per hour.

Day	Time	North Wall (W/m^2)	South Wall (W/m^2)	East Wall (W/m^2)	West Wall (W/m^2)	Roof Wall (W/m^2)	Total Radiation (W/m^2)
22 March	10:00 AM	453.52	149.03	149.03	395.60	677.83	1825.00
	11:00 AM	723.50	142.28	152.43	142.28	808.40	1968.89
	12:00 PM	793.87	140.93	140.93	309.79	875.07	2260.59
	1:00 PM	765.64	154.51	312.56	154.51	877.85	2265.06
	2:00 PM	540.36	177.73	472.43	177.73	808.40	2176.66
	3:00 PM	337.79	193.90	408.03	193.90	669.50	1803.12
	4:00 PM	221.34	186.05	273.39	186.05	488.93	1355.76
	5:00 PM	134.20	131.07	144.62	131.07	269.47	810.41
3 April	10:00 AM	446.71	151.97	151.97	531.94	755.62	2038.21
	11:00 AM	777.83	137.76	137.76	383.46	875.07	2311.87
	12:00 PM	912.86	133.43	215.35	133.43	936.19	2331.25
	1:00 PM	829.68	146.94	409.02	146.94	933.41	2466.00
	2:00 PM	722.90	174.32	231.98	174.32	866.74	2170.25
	3:00 PM	396.51	199.98	453.35	199.98	738.95	1988.78
	4:00 PM	259.33	209.73	332.50	209.73	580.60	1591.88
	5:00 PM	183.46	159.47	163.27	159.47	341.69	1007.37
4 April	10:00 AM	447.73	150.90	150.90	533.57	755.62	2038.73
	11:00 AM	780.63	136.54	136.54	383.79	875.07	2312.58
	12:00 PM	916.27	132.14	214.55	132.14	936.19	2331.28
	1:00 PM	832.67	145.65	409.37	145.65	933.41	2466.75
	2:00 PM	725.57	173.09	231.16	173.09	866.74	2169.66
	3:00 PM	397.18	198.91	454.52	198.91	738.95	1988.46
	4:00 PM	259.08	208.85	333.18	208.85	580.60	1590.56
	5:00 PM	183.49	158.94	162.83	158.94	341.69	1005.88

8. Embodied Energy, Energy payback time (EPBT), CO_2 mitigation and carbon credit earned are calculated.

4 | THERMAL MODELING OF EAHE

The following assumptions are considered for thermal modeling of EAHE:

1. The soil is isotropic, meaning that the soil properties do not change for different directions. This means that its conductivity will remain constant throughout the depth.
2. Because of the pipe's extremely thin thickness, it has very little heat resistance. It means that the pipe's surface temperature is equivalent to the soil temperature.
3. The temperature of the soil is the yearly average temperature of New Delhi which is 25°C and is constant throughout the depth of approximately 5 m.
4. The PVC pipe used is of uniform cross-sectional area and has a smooth inner surface, ensuring no slip condition.

Thermal modeling equations for the EAHE, Cengel and Boles[24]

i. Cross-sectional area is given by:

$$A_c = \frac{\pi D^2}{4} \tag{1}$$

where D is the diameter of PVC pipe.

ii. The curved surface area of the pipe where the heat transfer would take place is given by:

$$A_s = \pi \mathrm{DL} \tag{2}$$

iii. The mass flow rate of air is given by:

$$\dot{m} = \rho v A_c \tag{3}$$

where the density of air, $\rho = 1.204\,\mathrm{kg/m^3}$ and v is the velocity of air inside the EAHE.

iv. Convective heat transfer coefficient inside the pipe is given by:

$$h = \frac{N_u K}{D_h} \tag{4}$$

where N_u is the Nusselt's Number, K is the thermal conductivity, $K = 0.025\mathrm{W/mK}$ and D_h is the hydraulic length which in a circular tube is equal to the diameter of the pipe.

v. Nusselt's Number:
For a smooth and circular tube, it is given by the Gnielinski[25] for the following conditions:

$$N_u = \frac{\left(\frac{f}{8}\right)(R_e - 1000)Pr}{1 + 12.7\left(\frac{f}{8}\right)^{0.5}\left(Pr^{\frac{2}{3}} - 1\right)} \tag{5}$$

$$0.5 \le Pr \le 2000; 3 \times 10^3 \le R_e \le 5 \times 10^6$$

where f is the friction factor, Pr is the Prandtl number, and R_e is the Reynolds number.

vi. Friction factor, for a smooth pipe is given by Petukhov (1970) equation for the following condition:

$$f = (0.790 \ln(Re) - 1.64)^{-2} \tag{6}$$

$$3 \times 10^3 \le \mathrm{Re} \le 5 \times 10^6$$

vii. Prandtl Number is given by

$$\mathrm{Pr} = \frac{\rho v_k c_p}{K} \tag{7}$$

where $v_k = 1.51 \times 10^{-5}\,\mathrm{m^2/s}$, is the kinematic viscosity of air at 25°C, c_p is the specific heat of air at, 25°C $c_p = 1005\,\mathrm{J/kgK}$.

viii. NTU (number of transfer units) is given by:

$$NTU = \frac{hA_s}{\dot{m}c_p} \tag{8}$$

ix. Now, the Exit temperature can be determined from the equation:

$$T_e = T_s - (T_s - T_i)e^{-NTU} \tag{9}$$

where $T_s = 27°\mathrm{C}$, is the constant soil temperature and T_i is the ambient inlet air temperature.

x. The theoretical rate of heat transfer (Q_u) is given by using LMTD method:

$$T_l = \frac{T_i - T_e}{\log((T_s - T_e)/(T_s - T_i))} \tag{10}$$

$$Q_u = h \times A_s \times T_l \tag{11}$$

Energy balance equations for the room + EAHE system:

i. Absorbed Radiation Energy:

$$\alpha\left(\frac{I_{total}}{3600}\right)A = \frac{kA}{t}(T_{win} - T_{wout}) + AU_o(T_{wout} - T_a) \tag{12}$$

where $\frac{1}{U_o} = \frac{1}{h_o} + \left(\frac{t}{k}\right)5 + \left(\frac{1}{h_r}\right)4$. $h_o = 5.7 + 3.8 \times v$, Heat transfer coefficient of air outside the room. $h_r = 2.8 + 3 \times v$, Heat transfer coefficient of air inside the room. t = Thickness of the wall, 6 in = 0.15 m.

ii. Conduction of the absorbed energy:

$$\frac{kA}{t}(T_{win}-T_{wout})=h_rA(T_r-T_{win}) \quad (13)$$

iii. Convection of the conducted energy through the wall:

$$hA(T_r-T_{win})+Q_u=Mc_p\frac{dT_r}{dt}+\rho VNc_p(T_r-T_a)+UA(T_r-T_a) \quad (14)$$

where $\frac{1}{U}=\left(\frac{1}{h}\right)\times 5$.

$$\text{Number of air exchanges per second},N=\frac{A_c\times v}{V}.$$

T_{win} and T_{wout} are the effective inner wall and outer wall temperatures. It is assumed that all the walls have the same effective outer and inner wall temperatures to simplify the modeling. The values of the effective T_{win} and T_{wout} obtained through SIMULINK hold no importance. The focus of the modeling is on the results of room temperature only. T_r is the room temperature. T_a is the ambient temperature of the environment. The unknowns in these three equations are T_{win}, T_{wout}, and T_r. These equations are modeled in the MATLAB R2024b and solved using Simulink to get the room temperature.

4.1 | Reduction in cooling load

The total cooling load for air conditioner if temperature 24°C is to be maintained is:

$$Q_{AC}=\rho V(N\times 3600)c_p(T_a-24) \quad (15)$$

Therefore, the reduction in cooling load will be:

$$Q_{red}=\rho V(N\times 3600)c_p(T_a-T_r) \quad (16)$$

The validation of the experimental results with theoretical results is done by evaluating the correlation coefficient (r) and the root mean square percent deviation (e).[26] For experimental results to have a strong correlation with the theoretical results, the correlation coefficient should be between 0.75 and 1. The root mean square percent deviation indicates the percent deviation of experimental results from the theoretical results. These are given by the following equations:

$$\text{Correlation coefficient}\,r=\frac{N\sum X_iY_i-(\sum X_i)(\sum Y_i)}{\sqrt{N\sum X_i^2-(\sum X_i)^2}\sqrt{N\sum Y_i^2-(\sum Y_i)^2}} \quad (17)$$

where $r > 0$ indicates a positive linear relationship. $r < 0$ indicates a negative linear relationship. $r = 0$ implies no linear relationship between two variables and X_i is the theoretical values and Y_i is the experimental values.

$$\text{Root Mean Square Percent deviation},\,e=\sqrt{\frac{\sum(e_i)^2}{N}} \quad (18)$$

where $e_i=\frac{X_i-Y_i}{X_i}\times 100$.

4.2 | Embodied energy

The total embodied energy of a system indicates the amount of energy needed to build that system. It considers all the factors

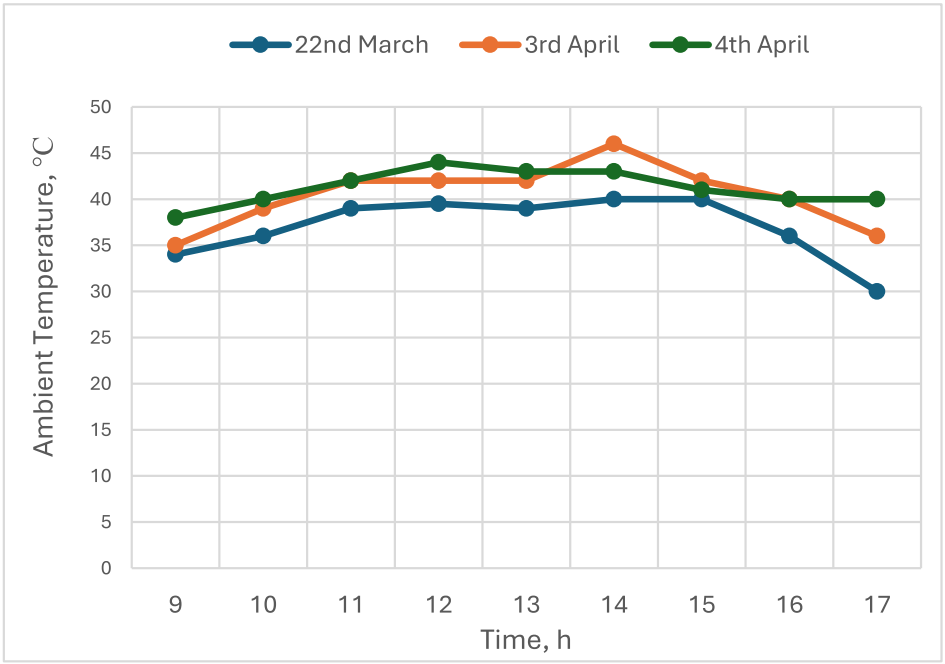


FIGURE 2 The hourly variation of ambient temperature on the experimental days.

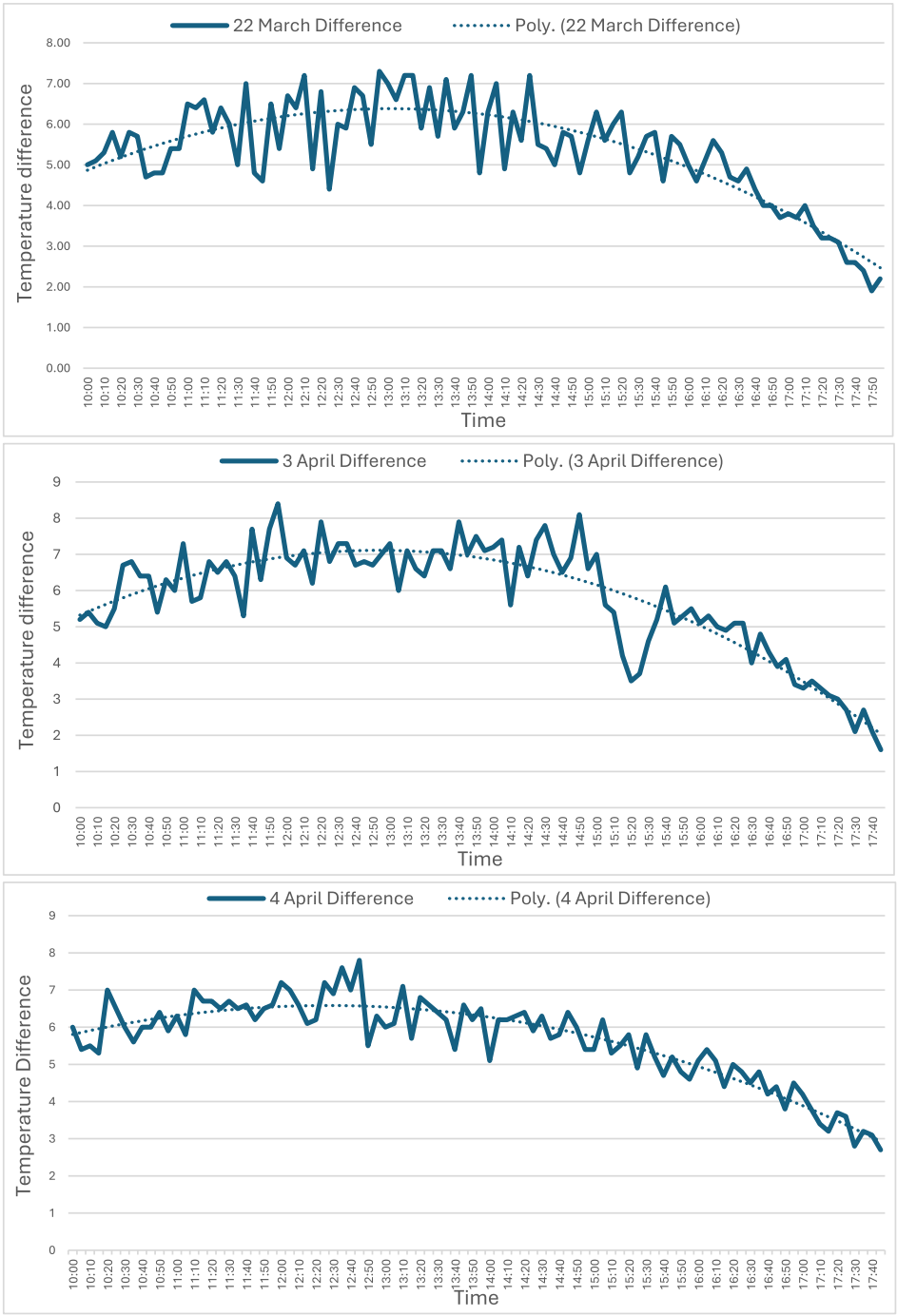


FIGURE 3 Temperature difference between the inlet and outlet air temperature for different days.

including the energy needed to extract, process, and transport the raw materials. The analysis of this embodied energy is needed to ensure that the invested energy is generated back to balance it.

4.3 | Energy payback time (EPBT)

The amount of time required to produce the energy equivalent to the total embodied energy is termed as EPBT. It is the ratio of total embodied energy to the total annual energy output.

$$\text{EPBT} = \frac{\text{Embodied Energy}, (E_{in}, \text{kWh})}{\text{Annual Energy Output}\,(E_{aout}, \text{kWh/year})} \quad (19)$$

4.4 | Net CO_2 emission and mitigation

The average CO_2 equivalent intensity for electricity generation from coal is approximately 0.98 kg of CO_2 per kWh at the source.

$$\text{Annual CO}_2\text{ emission per year} = \frac{E_{in}}{L} \times \frac{1}{1-L_a} \times \frac{1}{1-L_t} \times 0.98\,\text{kg} \quad (20)$$

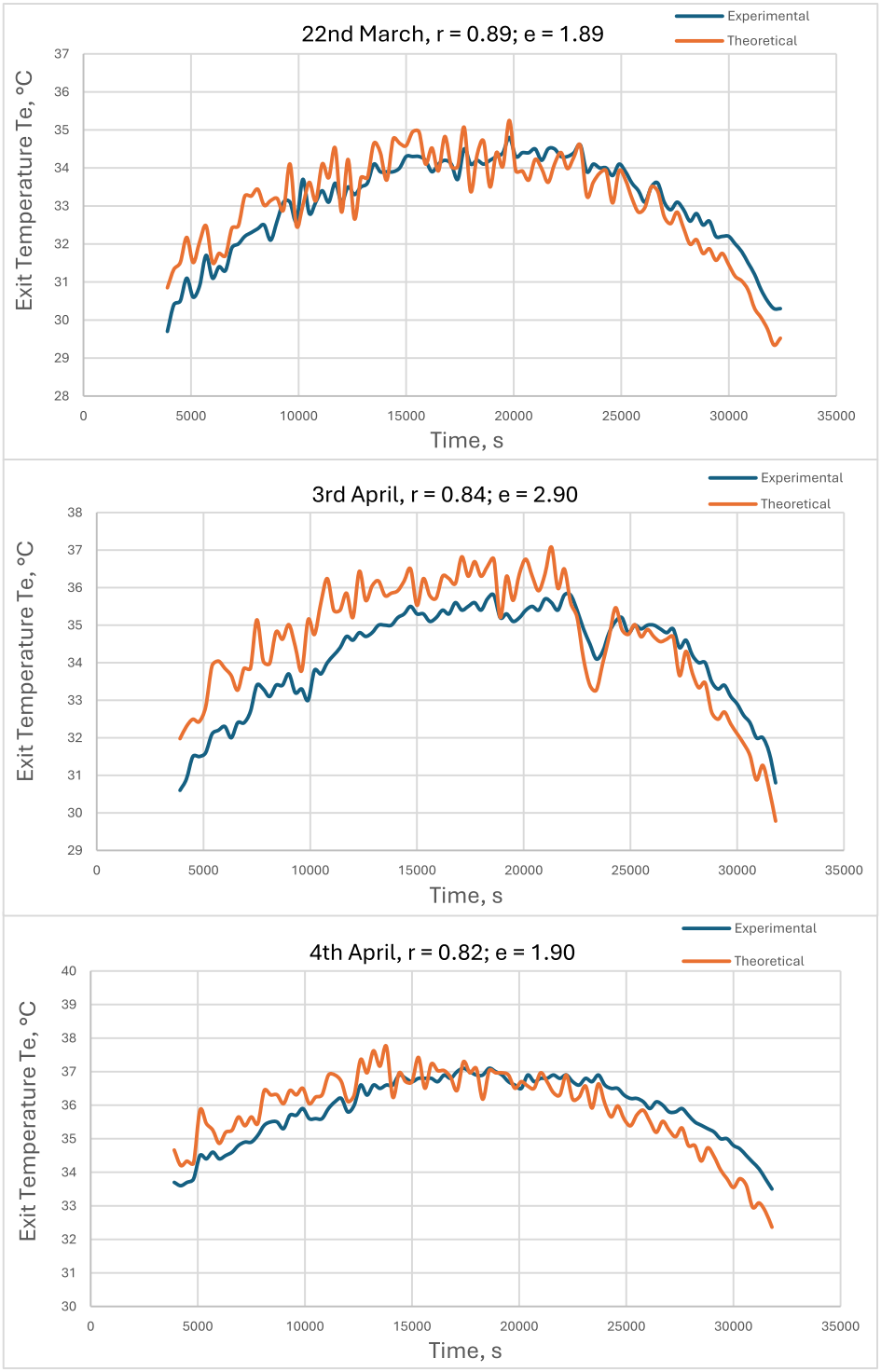


FIGURE 4 Variation of theoretical exit temperature with experimental validation for each day.

The transmission losses (L_t) = 0.4 and domestic appliances losses (L_a) = 0.2 according to R.K Mishra et al. and the assumed life span of EAHX (L) = 20 years

$$\text{Net } CO_2 \text{ mitigation} = (E_{aout} \times L - E_{in}) \times \frac{1}{1-L_a} \times \frac{1}{1-L_t} \times 0.98\,\text{kg} \quad (21)$$

4.5 | Carbon credit earned

If CO_2 emission is being traded at 10 USD per ton of CO2 mitigation, then the carbon credit earned by the system in USD is obtained as

$$\text{Carbon Credit Earned} = \text{Net } CO_2 \text{ mitigation} \times \$10 \quad (22)$$



FIGURE 5 Hourly variation of theoretical room temperature with experimental observation for each day.

5 | RESULTS AND DISCUSSION

The total hourly variation of solar radiation incident on different walls of the room on different days is shown in Table 1. The hourly ambient temperature has been measured using a mercury thermometer.

Figure 2 shows the hourly variation of the ambient temperature for each day. In the EAHE, the inlet and outlet temperatures are measured using two K-type thermocouples, which are installed at the inlet and outlet of the pipe. Also, the room temperature is measured hourly using a mercury thermometer. To prevent thermal lag and any errors due to transient conditions, initial readings of the exit temperature are neglected for 1 h so that steady-state conditions are established. Figure 3 shows the temperature difference between inlet and exit air temperatures for 3 days. On 22nd March, it is observed that the inlet temperature ranges from 32.2°C to 41.7°C. The temperature difference ranges from 1.9 to as high as 7.3. This difference corresponds to the lowest inlet temperatures of 32.2°C and 41.2°C, respectively, which is close to the highest temperature. A higher temperature

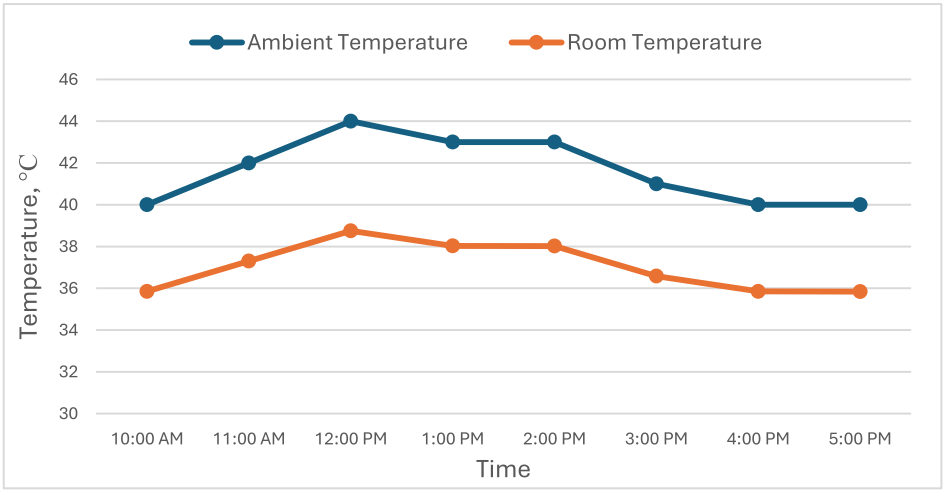


FIGURE 6 Hourly variation of ambient temperature and room temperature.

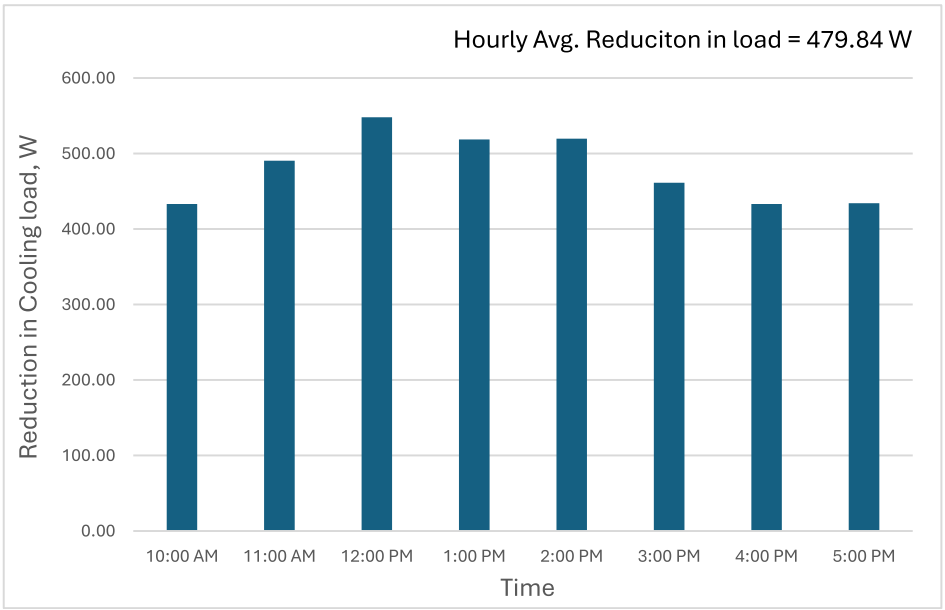


FIGURE 7 Hourly reduction in cooling load of the room.

TABLE 2 Calculation of embodied energy for the proposed EAHE.

Sr. no.	Material	Quantity (kg)	Embodied energy (kWh/kg)	Total embodied energy (kWh)
1	PVC	80	21.11	1681.50
2	Plastic fans (for 20 years of EAHE life cycle)	1 × 20	19.44	388.8
Total embodied energy				2070.30

difference indicates a higher rate of heat transfer, or in other words, a higher rate of thermal energy. Similar observations were made on 3rd April and 4th April. Hence, it is correct to say that when the temperature difference between the inlet and the soil temperature (which in this case is 25°C), is more than the temperature difference between the inlet and the exit is more which corresponds to a higher rate of heat transfer.

5.1 | Validation of the EAHE thermal modeling

The experimentally measured inlet air temperature over 8 h was used as input to the thermal model, and the corresponding theoretical outlet temperatures were obtained for all 3 days. Also, the air flow velocity in the thermal modeling was changed according to the setup. Figure 4 shows the plot between theoretical and experimental exit temperature for each day. Root mean square percent deviation (e) and correlation coefficient (r) have been calculated for the same. It can be observed that the theoretical model is in good agreement with the experimentally obtained values. The correlation coefficients are between 0.84 and 0.89, and the RMS percent deviation is less than

TABLE 3 CO_2 emissions per year for different life spans of EAHE.

Lifespan of EAHE (years)	10	15	20	50
CO_2 emission (kg/year)	422.69	281.79	211.34	84.54

TABLE 4 CO_2 mitigation (tons) for different life spans of EAHE.

Lifespan of EAHE (years)	10	15	20	50
CO_2 mitigation (in ton)	5.32	10.10	14.87	43.51

3%, which shows the percent deviation is minimal. Hence, the theoretical model results are validated.

5.2 | Validation of room EAHE system

The hourly total solar radiation incident on the wall of the room and the hourly ambient temperature which is obtained experimentally is fed to the thermal modeling of room + EAHE system. The other parameters and constants used in modeling are given in Table A1. The theoretical room temperature is obtained and compared to the experimental room temperature measured hourly using a mercury thermometer. Figure 5 shows the plot between theoretical and experimental room temperature for 3 days. Root mean square percent deviation and correlation coefficient have been calculated for the same.

It can be observed that the theoretical model is in good agreement with the experimental obtained values. Strong correlation coefficients, close to 0.85, are obtained for each day having different air flow velocity. The RMS percent deviation is less than 4%, which shows the percent deviation is minimal. Hence, the theoretical model results are validated.

5.3 | Reduction in cooling load for summer season

Now that the theoretical model has been validated, the cooling load calculation is considered for a realistic size of room (12 ft. × 12 ft. × 12 ft/8.72 m × 8.72 m × 8.72 m). Summer season is considered of 4 months (March, April, May and June). Cooling load reduction is calculated for a single day (4 April) and has been extrapolated to 4 months, that is, 122 days.

This is good assumption since the lower temperature difference between ambient and soil in March is compensated by higher temperature difference in May. Hence, it is safe to take the reduction in cooling load for April and extrapolate to the entire summer season. All the conditions in the modeling which are air flow velocity (1.2 m/s), ambient temperature and solar radiation incident, remains the same as it was on 4 April, Figure 6 shows the ambient temperature and new room temperature obtained hourly on 4 April after considering the realistic size of room. Figure 7 shows the hourly reduction in cooling load of the air conditioner. It is found that on an average 479.84 W of cooling load is reduced due to the integration of EAHE. This reduction corresponds to daily saving of 3.8 kWh of power if the system works for 8 h. For the entire summer season, 468 kWh of electricity is saved after integrating EAHE to the room.

TABLE 5 Carbon credit earned for different life spans of EAHE.

Lifespan of EAHE (years)	10	15	20	50
Carbon Credit Earned ($)	$53.24	$100.97	$148.71	$435.13

5.4 | Energy payback time

The total embodied energy needed to construct this system is shown in Table 2. The annual energy output per year is the total amount of reduction in cooling load. Considering the summer season, it is found that 468 kWh of energy is reduced annually.

$$\text{EPBT} = 2075.20/468 = 4.42 \text{ years}.$$

5.5 | Carbon credit

The CO_2 emissions, CO_2 mitigation, and the carbon credit earned during different lifespan of the EAHE are shown in Tables 3, 4, and 5, respectively.

6 | CONCLUSION

Based on this study, the following conclusions were made:

- Temperature difference between the inlet and exit temperatures increases as the temperature difference between the ambient air temperature and constant soil temperature increases.
- The validation of the experimental results with both the modeling in MATLAB showed strong correlations, r ranging between 0.82 and 0.89 and low root mean square percent deviation, e ranging between 2.17 and 4.84, confirming both models' accuracy in predicting the exit temperature as well as the room temperature.
- The energy consumption of AC without integrating EAHE, running for 9 h a day, is found to be 14.72 kWh. The total reduction in cooling load on 4 April for a room of 1.2 ft × 12 ft × 12 ft is found to be 3.2 kWh and for the entire summer season, it is found to be 468 kWh reduction in cooling load. Thus, integrating the EAHE system with the AC unit will reduce the cooling load by 20%.
- The EPBT of the room + EAHE system is found to be 4.42 years when the EAHE works in the summer season. CO_2 mitigation and carbon credit earned for 50 years of EAHE are found to be 43.50 Ton and $ 435, respectively, which contributes to environmental sustainability.

The experimental validation of this theoretical model for passive heating during the winter season should be performed. After this

validation, the total reduction in the heating load can be assessed, which would be highly informative for evaluating the total reduction in cooling and heating loads throughout the year. Furthermore, EPBT and carbon mitigation can be calculated, which is valuable for assessing its environmental sustainability.

NOMENCLATURE

A_c	Cross sectional area of pipe (m^2)
A_s	Curved surface area of pipe (m^2)
A	Area of the wall of room (m^2)
D	Pipe diameter (m)
e	Root mean square percent deviation
$\dot{m}$	Mass of flow rate inside the EAHE (kg/s)
M	Mass of the air inside room (kg)
L	Length of pipe (m)
r	Correlation coefficient
V	Volume of room (m^3)
v	Velocity of air (m/s^2)
N_u	Nusselt's Number
D_h	Hydraulic length (m)
K	Thermal conductivity of PVC (W/mK)
k	Thermal conductivity of concrete wall (W/mK)
f	Friction factor
R_e	Reynolds Number
Pr	Prandtl Number
v_k	Kinematic viscosity of air (m^2/s)
c_p	Specific heat of air (J/kgK)
NTU	Number of transfer unit
T_e	Exit temperature of air (°C)
T_i	Inlet temperature of air (°C)
T_s	Soil temperature (°C)
T_{win}	Effective inner wall temperature of the room (°C)
T_{wout}	Effective outer wall temperature of the room (°C)
T_a	Ambient temperature (°C)
T_r	Room temperature (°C)
h	Heat transfer coefficient inside the pipe (W/m^2K)
h_r	Heat transfer coefficient inside the room (W/m^2K)
h_o	Heat transfer coefficient outside room (W/m^2K)
I_{Total}	Total intensity of solar radiation incident on all the walls of the room including the roof per hour (W/m^2h)
t	Thickness of wall (m)
U_o	Heat loss coefficient outside the room (W/m^2K)
U	Heat loss coefficient inside the room (W/m^2K)
N	Number of air exchanges per second (s^{-1})
Q_{AC}	Total Cooling load for Air conditioner without EAHE (W)
Q_{red}	Reduction in cooling load of the room due to EAHE (W)
ρ	Density of air (kg/m^3)
α	Absorptivity of concrete

AUTHOR CONTRIBUTIONS

Harender Sinhmar: Funding acquisition; writing – original draft; writing – review and editing; project administration; supervision; validation. **Devarshi Patel:** Investigation; writing – original draft; writing – review and editing. **Ravinder Sahdev:** Conceptualization; writing – original draft; writing – review and editing; supervision; visualization; investigation. **Rajeev Kumar Mishra:** Supervision; writing – original draft; writing – review and editing; investigation; validation; formal analysis; visualization. **Sumit Tiwari:** Conceptualization; writing – original draft; writing – review and editing; supervision; resources; funding acquisition; validation; visualization; project administration. **Arpit Patel:** Investigation; validation; software; formal analysis; data curation; writing – review and editing; writing – original draft.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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APPENDIX A

TABLE A1 Variation of design parameters taken for calculation of different temperatures.

$\rho = 1.204\,\text{kg/m}^3$	$T_s = 27^\circ\text{C}$	$k = 0.5\,\text{W/mK}$	$L = 9.1\,\text{m}$
$K = 0.025\,\text{W/mK}$	$\alpha = 0.3$	$M = 2.08\,\text{kg}$	$A_c = 0.071\,\text{m}^2$
$v_k = 1.51 \times 10^{-5}\,\text{m}^2/\text{s}$	$v = 0.5, 1, 1.2\,\text{m/s}$	$V = 1.728\,\text{m}^3$	$A_s = 8.576\,\text{m}^2$
$c_p = 1005\,\text{J/kgK}$	$t = 0.15\,\text{m}$	$A = 1.44\,\text{m}^2$	$D = 0.3\,\text{m}$